Threat Model: N-Agent Mesh

Adversary
no model or key access

Write channels
User query · Tool / API
Peer agent · Self-written

poisoned entry

Agent 1 — false fact — Memory store
valid signature, but the agent was misled

forward, hop 1

Agent 2 — copy — Memory store
copies reach neighbors

summarize, hop 2

Agent 3 — summary copy — Memory store
summaries inherit poison

via 4…N−1

Agent N — reworded copy — Memory store
reworded copy misleads later

Agentic IoT applications
IoT Agents with Shared Memory

Connected vehicles
route and traffic memory

UAV fleets
mission and map memory

Industrial IoT
maintenance logs

Smart city
gateway service memory

Proposed *EngraMesh* Framework

New entry from any channel

Runs locally on the edge devices

Layer 1
Signed Envelope

Memory entry
- Writer ID
- Write channel
- Parent IDs
- Trust label, expiry
- Content hash

Ed25519 signed

Merkle log · **Verify**

Layer 2
Lightweight Detector

Stage 1: Text classifier

Entry text → MiniLM embedding → MLP classifier → Content score

Stage 2: Fusion

Provenance
channel, reputation, lineage

Network
hops, holders, spread

Fusion (LR)

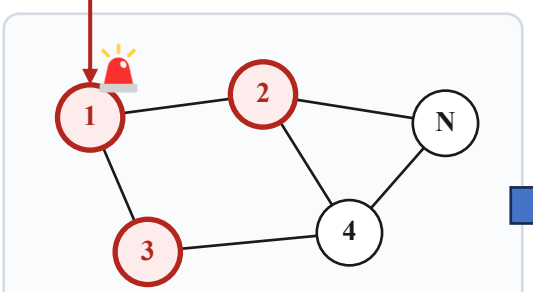
Layer 3
Accept, Quarantine, or Recall

Risk score *s*

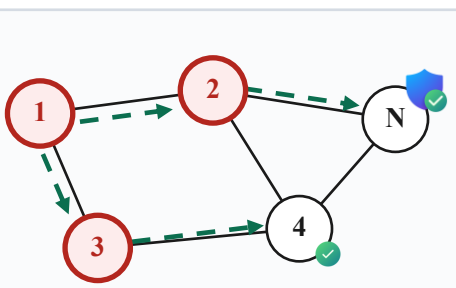
- $s < 0.3$ — **Accept** — Kept in memory
- $0.3 \le s < 0.7$ — **Quarantine** — Held for review
- $s \ge 0.7$ — **Recall** — Removed and sent to agents

RECALL(entry, lineage)
signed, also removes near-copies

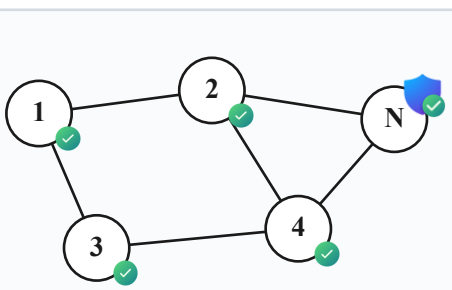
Network-Wide Recall
Removing the poisoned entry across the mesh



Agent 1 detects the poisoned entry when it is used



Signed RECALL hops along other agents and removes near-copies



Mesh cleared and late copies blocked

Legend
- holds the poison
- clean agent
- RECALL hop
- flags the entry
- blocks late copies
- inside a device
- between agents